

Literature Mining Report

Query Statistics

Query Date: 2025-12-19

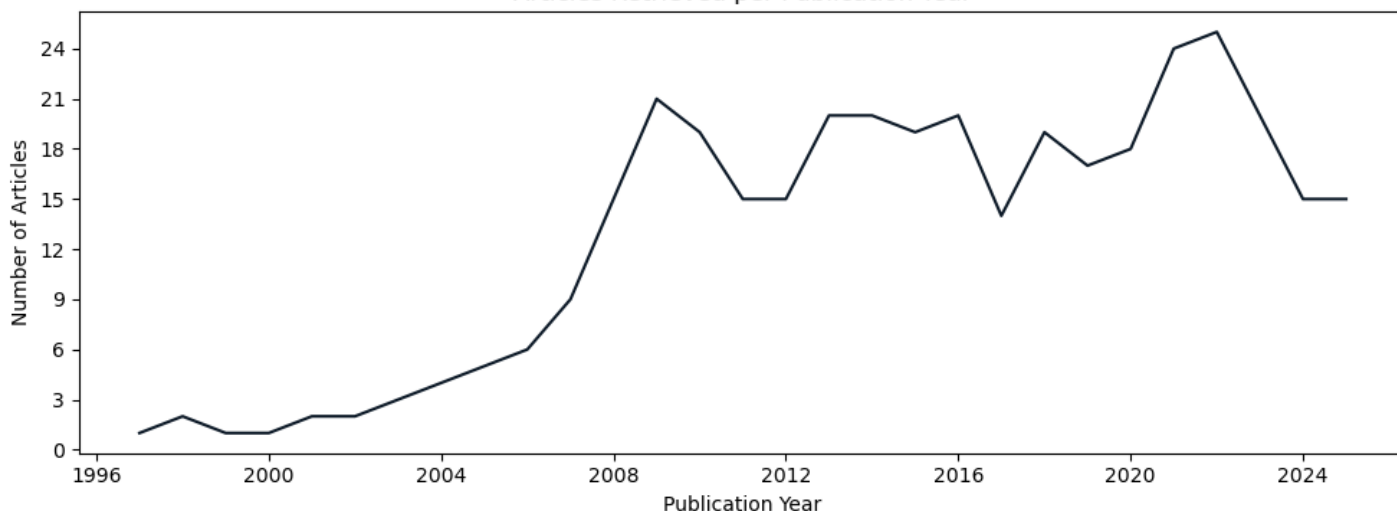
Query Terms: "ASE asymmetry c elegans"

Scoring Keywords: che-1; Isy-6 (Strict Mode)

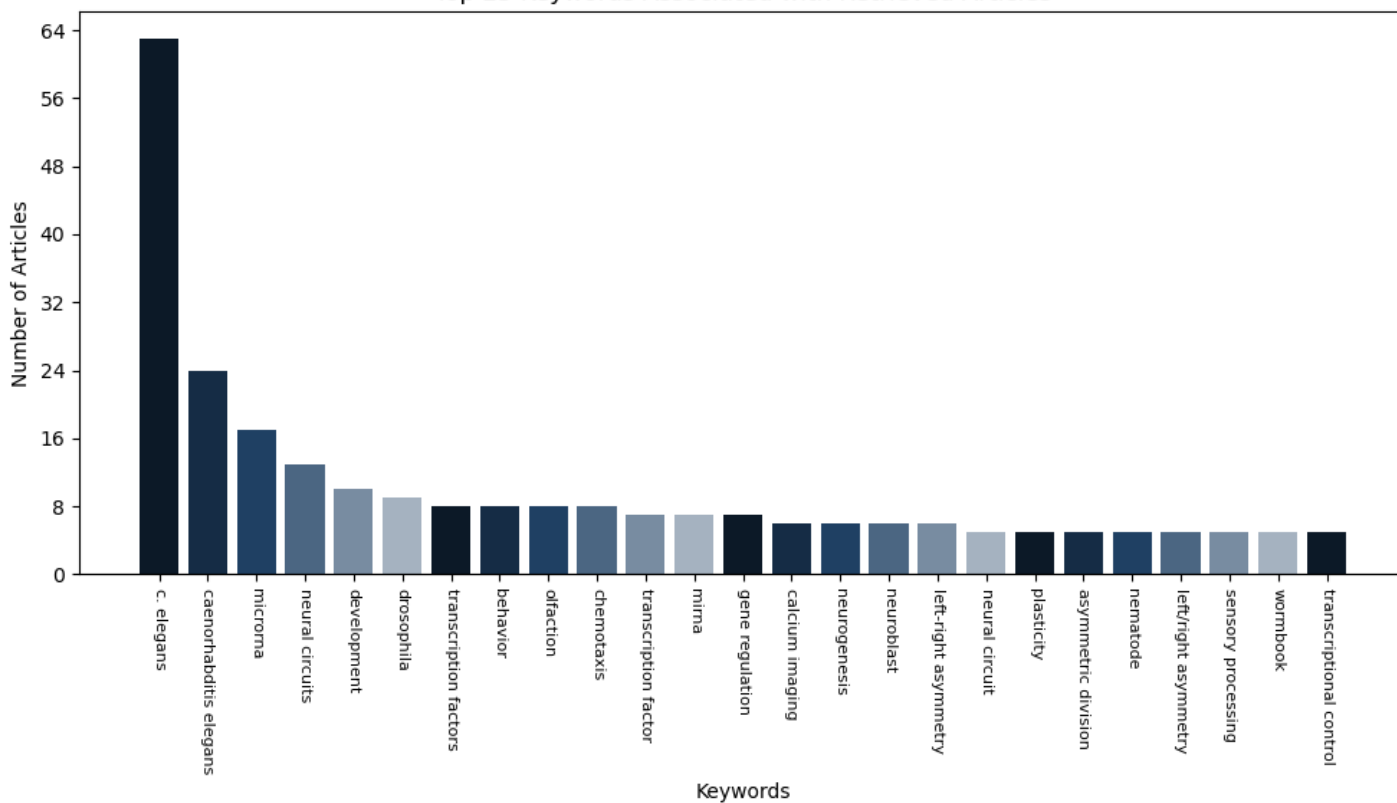
Articles Parsed: 371/371 (100.00%)

Keyword Rarity Score (IDF): Isy-6: 2.515; che-1: 2.179

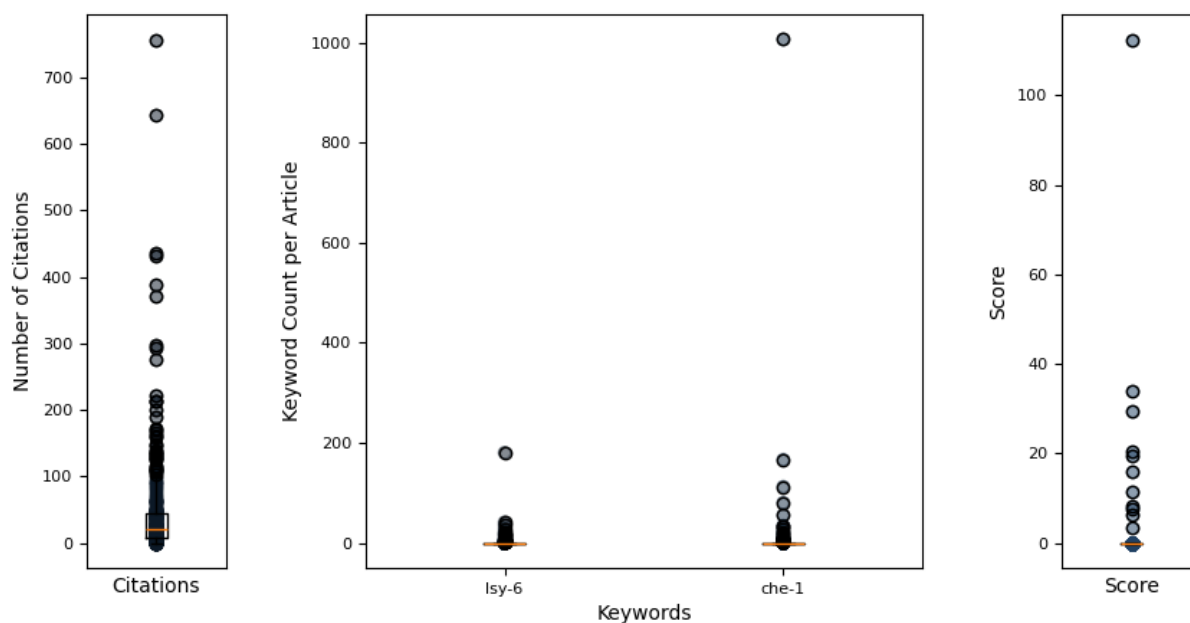
Articles Retrieved per Publication Year



Top 25 Keywords Associated with Retrieved Articles



Hermes Report Summary



Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
33002421	2020	Developmental C	Charest, Julien et al.	combinatorial action of temporally segregated...	27	112.52
40586702	2025	eLife	NA	evolution of lateralized gustation in nematod...	2	33.78
39282255	2025	bioRxiv	Mackie, Marisa et al.	evolution of lateralized gustation in nematod...	0	29.51
24065887	2013	Frontiers in Ce	Alqadah, Amel et al.	microRNA function in left-right neuronal asym...	11	20.35
21698137	2011	PLoS Genetics	Poole, Richard et al.	a genome-wide rna screen for factors involve...	33	19.57
31526477	2019	eLife	Hong, Ray L et al.	evolution of neuronal anatomy and circuitry i...	35	15.88
19875417	2010	Nucleic Acids R	Takayama, Jun et al.	single-cell transcriptional analysis of taste...	18	11.43
26635721	2015	Frontiers in Ne	Davis, Gregory et al.	micrornas: not fine-tuners but key regulato...	42	8.26
29481550	2018	Nature methods	Alberti, Chiara et al.	cell-type specific sequencing of micrornas fr...	51	7.60
39167071	2024	Genetics	Poole, Richard et al.	Neurogenesis in Caenorhabditis elegans	11	6.30
34504291	2021	Communications	Dekkers, Martij et al.	plasticity in gustatory and nociceptive neuro...	8	3.24

Query Result 1/11**Combinatorial Action of Temporally Segregated Transcription Factors**

Charest, Julien; Daniele, Thomas; Wang, Jingkui; Bykov, Aleksandr; Mandlbauer, Ariane; Asparuhova, Mila; Rhsner, Josef; Gutierrez-Prez, Paula; Cochella, Luisa

Developmental Cell

Year: 2020 PMID: 33002421 Citations: 27 Score: 112.52 DOI: 10.1016/j.devcel.2020.09.002

Keywords Hits: Isy-6: 180, che-1: 55

Keywords Frequency (TF): Isy-6: 0.014; che-1: 0.004

Abstract:

Combinatorial action of transcription factors (TFs) with partially overlapping expression is a widespread strategy to generate novel gene-expression patterns and, thus, cellular diversity. Known mechanisms underlying combinatorial activity require co-expression of TFs within the same cell. Here, we describe the mechanism by which two TFs that are never co-expressed generate a new, intersectional expression pattern in *C.elegans* embryos: lineage-specific priming of a gene by a transiently expressed TF generates a unique intersection with a second TF acting on the same gene four cell divisions later; the second TF is expressed in multiple cells but only activates transcription in those where priming occurred. Early induction of active transcription is necessary and sufficient to establish a competent state, maintained by broadly expressed regulators in the absence of the initial trigger. We uncover additional cells diversified through this mechanism. Our findings define a mechanism for combinatorial TF activity with important implications for generation of cell-type diversity.

Summary:

Introduction: TFs are a key component of the development of neurons. Combinatorial use of TF enables the generation of novel, specific gene-expression patterns that exceed the number of available TF. The specification of individual cell identities by combinations of CFs is a basis for cellular diversification during development. The concept of temporal intersection in which two TF are required for transcriptional activation, but separated in time. TF1 is expressed transiently in a subset of progenitor cells, while TF2 is expressed in a specialized subset that is undergoing differentiation.

Results: Direct, early binding of TBX-37/38 is necessary and sufficient to establish competence for transcription in the post-mitotic ASEL Neurons. To begin to dissect the mechanism by which TBX-37 and tbx-38 contribute to the specific expression of Isy-6 in the post mitotic ASE, we explored the possibility that these TFs bind directly to the Isy-6 locus already at the 28-cell stage, when they are first detected.

Discussion: We show that two temporally segregated TFs act combinatorially to achieve cell-type specific transcription of a locus that controls neuronal identity. We also provide evidence that temporal integration of TF may be a more general mechanism for cellular diversification. The use of TBX-37/38 for gene activation is a widespread phenomenon that has been long recognized to enable generation of cell-type-specific gene-expression patterns and, in turn, cell-type diversity. Here, we revealed mechanistic insight into how two temporally segregate TF combinations can act combinatorially to achieve cellular specific transcription.

Figures:

Figure 1: Integration of Transcriptional Inputs over Time as a Mechanism for Cell Diversification

Figure 2: Early, Direct Binding of TBX-37/38 to Isy-6 Is Necessary and Sufficient to Establish Competence for Transcription in the ASE Neurons

Figure 3: TBX-37/38 Are Not Continuously Required for Isy-6 Expression

Figure 4: TBX-37/38 Establish a Differentially Accessible State of the Isy-6 Locus

Figure 5: Transcription over the Isy-6 Locus Occurs Bidirectionally and Is Required for Priming

Figure 6: Asymmetric Isy-6 Competence Is Maintained in a Symmetric Trans-acting Factor Environment

Figure 7: TBX-37/38 Determine Left-Right Molecular Asymmetry across Multiple Bilateral Neuron Pairs with ABa/ABp Lineage Origin

Mentioned biomedical entities:

Genes: ASE, ASER, AVE, CHE-1, Cas9, FoxA, Mos1, P40, PHA-4, SEP, SM2, TBX-37, TBX-38, TEL, TF, TF1, VP16, ZIF-1, Zdbf2, cat, galK, gcy-5, histone, Isy-6, unc-119

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell, neuron

Organisms: E. coli, E.coli, cat

Tissues: capillary, egg, eggshell, neuron, tube

Pathways: N/A

Query Result 2/11

Evolution of lateralized gustation in nematodes

NA

eLife

Year: 2025 PMID: 40586702 Citations: 2 Score: 33.78 DOI: 10.7554/eLife.103796.3.sa0

Keywords Hits: lsy-6: 7, che-1: 112

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.012

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in *Pristionchus pacificus* respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode *Caenorhabditis elegans*. Previously we showed that *P. pacificus* likely lacked bilateral asymmetry (Hong et al., 2019). Here, we show that by visualizing neuronal activity patterns, contrary to previous expectations based on its genome sequence, the salt responses of *P. pacificus* are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Introduction: Nematodes form a vast array of ecological relationships, from specialized parasite-host dependencies to nematode-microbial interactions, each one demanding exquisitely fine-tuned sets of sensory palates that span multiple modalities. Yet, the number of sensory neurons across diverse nematode species seems to be constrained. To identify the genetic basis for patterning different nervous systems and to understand the processes that underlie evolutionary changes in adapting to different environments, several comparative model systems have been developed to promote comparisons to the well-studied nematode *Caenorhabditis elegans* at the genetic and cellular levels, including the predatory entomophilic nematode, *Pristionchus pacificus*.

Results: Comparing the chemosensory profiles of *P. pacificus* and *C. elegans* suggests differences in their salt preferences. NH₄I is more attractive to *P. pacificus* than *C. elegans*. Similarly, *P. pacificus* is less attracted to NaCl and LiCl than to ammonium salts. Interestingly, *P. pacificus* is repulsed by acetate salts, which induce attractive responses in *C. elegans*.

Discussion: Our study has revealed several insights into the substrates of evolutionary changes between two distantly related nematode species, *P. pacificus* and *C. elegans*. We used intracellular calcium levels as our readout for neuronal activity with a genetically encoded calcium sensor in two pairs of *che-1*-expressing amphid sensory neurons the first calcium imaging study in *P. pacificus*. We show that three neuron types (ASE left, ASE right, and AFD neurons) each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. We have identified the first laterally asymmetric marker in *P. pacificus*, *gcy-22.3p::GFP*, with its expression limited to ASER homolog (AM7). Our unexpected discovery that neuron asymmetry is present in the ASE homologs between two closely related nematodes, despite the lack of an *alsy-6* homolog, suggests that functional lateralization in *P.*

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions between *P. pacificus* and *C. elegans*.

Figure 2: *P. pacificus che-1* expression in ASE and AFD amphid neurons.

Figure 3: Co-expression of *gcy-22.3p::GFP* and *txx-1p::RFP* in ASER.

Figure 4: *che-1* expressing amphid neurons are required for sensing water-soluble ions in *P. pacificus*.

Figure 5: The *che-1p::HisCl1* transgene is necessary for histamine-dependent knockdown of salt attraction.

Figure 6: Molecular lesions for *P. pacificus che-1* and *gcy 22.3*.

Figure 7: ASER and ASER responses to different concentrations of NH₄Cl, NaCl, and NH₄I.

Figure 8: Negative controls of ASE and AFD neuron responses to green light.

Figure 9: Individual traces and heatmaps of ASE responses to various salts.

Figure 10: Microfluidic apparatus used for calcium imaging.

Figure 11: Combined AFD responses in comparison to ASE left or right neuron responses to NH₄Cl, NaCl, and NH₄I.

Figure 12: Individual traces and heatmaps of AFD responses to various salts.

Figure 13: AFDL/R responses to different concentrations of NH₄Cl, NaCl, and NH₄I.

Figure 14: The laterally asymmetric expression of *gcy-22.3* is dependent on the zinc finger transcription factor CHE-1.

Figure 15: ASE and AFD responses in wildtype compared to *gcy-22.3* mutants.

Figure 16: AFDL and AFDR responses to NH₄Cl and NaCl in wildtype and *gcy-22.3* mutant.

Figure 17: Chemotaxis responses to individual ions in *gcy-22.3* mutants.

Mentioned biomedical entities:

Genes: ADF, ALFA, ASE, ASER, ASK, B, CHE-1, Cas9, Cel, HisCl1, IMD, MA, Max, Notch, OTX, Ppa, RIP, SHC, TTX-1, Twist, *cog-1*, *die-1*, *fozi-1*, *gcy*, *gcy-22*, *gcy-8*, *lim-6*, *lin-4*, *lsy-6*, *ttx-1*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: *C. elegans*, human

Tissues: neuron, ventral

Pathways: N/A

Query Result 3/11

Evolution of lateralized gustation in nematodes

Mackie, Marisa; Le, Vivian Vy; Carstensen, Heather R.; Kushnir, Nicole R.; Castro, Dylan L.; Dimov, Ivan M.; Quach, Kathleen T.; Cook, Steven J.; Hobert, Oliver; Chalasani, Sreekanth H.; Hong, Ray L.

bioRxiv

Year: 2025 PMID: 39282255 Citations: 0 Score: 29.51 DOI: 10.1101/2024.08.31.610597

Keywords Hits: lsy-6: 5, che-1: 80

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.013

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in *Pristionchus pacificus* respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode *Caenorhabditis elegans*. By visualizing neuronal activity patterns, we show that contrary to previous expectations based on its genome sequence, the salt responses of *P. pacificus* are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Results: *P. pacificus* and *C. elegans* display differences in their salt preferences. Previous cross-species comparisons between *P. eliotus* and *C. elegans* indicated strong differences in olfaction preferences that reflect their divergent evolutionary histories and ecology. To identify the neurons that mediate gustation, we first compared the chemosensory profiles of these two species toward water-soluble ions. In this survey, we found ammonium salts to be the strongest attractants to wildtype *P. pacificus*.

Discussion: Our study has revealed several insights into the substrates of evolutionary changes between two distantly related nematode species, *P. pacificus* and *C. elegans*. We used intracellular calcium levels as our readout for neuronal activity with a genetically encoded calcium sensor in two pairs of *che-1*-expressing amphid sensory neurons the first calcium imaging study in *P. pacificus*. We show that three neuron types (ASE left, ASE right and AFD neurons) each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. We have identified the first laterally asymmetric marker in *P. pacificus*, *gcy-22.3p* GFP, with its expression limited to ASER homolog (AM7). Our unexpected discovery that neuron asymmetry is present in the ASE homologs between two closely related nematodes, despite the lack of a *lsy-6* homolog, suggests that functional lateralization in *P. pacificus*.

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions between *P. pacificus* and *C. elegans*.

Figure 2: *P. pacificus* *che-1* expression in ASE and AFD amphid neurons.

Figure 3: *che-1* expressing amphid neurons are required for sensing water-soluble ions in *P. pacificus*.

Figure 4: ASEL and ASER responses to different concentrations of NH₄Cl, NaCl, and NH₄I.

Figure 5: Combined AFD responses in comparison to ASE left or right neuron responses to NH₄Cl, NaCl, and NH₄I.

Figure 6: The laterally asymmetric expression of *gcy-22.3* is dependent on the zinc finger transcription factor CHE-1.

Figure 7: ASE and AFD responses in wildtype compared to *gcy-22.3* mutants.

Figure 8: Chemotaxis responses to individual ions in *gcy-22.3* mutants.

Mentioned biomedical entities:

Genes: ADF, ALFA, ASE, ASER, ASK, CA, CHE-1, Cas9, Cel, IMD, MA, Notch, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy, lim-6, lin-4, lsy-6, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: *C. elegans*

Tissues: neuron, ventral

Pathways: N/A

Query Result 4/11microRNA function in left-right neuronal asymmetry: perspectives from *C. elegans*

Alqadah, Amel; Hsieh, Yi-Wen; Chuang, Chiou-Fen

Frontiers in Cellular Neuroscience

Year: 2013

PMID: 24065887

Citations: 11

Score: 20.35

DOI: 10.3389/fncel.2013.00158

Keywords Hits: lsy-6: 20, che-1: 3

Keywords Frequency (TF): lsy-6: 0.008; che-1: 0.001

Abstract:

Left-right asymmetry in anatomical structures and functions of the nervous system is present throughout the animal kingdom. For example, language centers are localized in the left side of the human brain, while spatial recognition functions are found in the right hemisphere in the majority of the population. Disruption of asymmetry in the nervous system is correlated with neurological disorders. Although anatomical and functional asymmetries are observed in mammalian nervous systems, it has been a challenge to identify the molecular basis of these asymmetries. *C. elegans* has emerged as a prime model organism to investigate molecular asymmetries in the nervous system, as it has been shown to display functional asymmetries clearly correlated to asymmetric distribution and regulation of biologically relevant molecules. Small non-coding RNAs have been recently implicated in various aspects of neural development. Here, we review cases in which microRNAs are crucial for establishing left-right asymmetries in the *C. elegans* nervous system. These studies may provide insight into how molecular and functional asymmetries are established in the human brain.

Summary:

INTRODUCTION: MicroRNAs (miRNAs) are endogenous 20-24 nt small non-coding RNAs that regulate gene expression through binding to complementary sequences in target messenger RNA (mRNAs), leading to translational repression and/or cleavage of target mRNA(s). While most miRNAs downregulate gene expression, there are examples of miRNA-mediated upregulation of target gene expression during cell cycle arrest, suggesting that miRNA function is complex and context dependent. miRNAs have been implicated in many aspects of development and disease including cell cycle, cell differentiation, apoptosis, life span, developmental timing, stress responses, neural development and regeneration, cancers, and neurodegenerative disorders.

PERSPECTIVES: The role of miRNAs in establishing left-right asymmetry in the *C. elegans* nervous system can lay the groundwork for identifying the processes used in higher organisms, as the methods used may be evolutionarily conserved. Because of the highly conserved nature of miRNAs, insights into how they are involved to control asymmetric fates will help facilitate our understanding in vertebrate neuronal asymmetry. The involvement of miRNAs in the asymmetric process may also be reflective of the principles these small non-coding RNAs use in directing other neurodevelopmental processes.

Figures:

Figure 1: Priming and boosting model of ASE asymmetry. ASE: (1) The first step of promoting the ASE cell fate occurs six cell divisions before the postmitotic ASE is born. A transient pair of T-box transcription factors TBX-37/38 acts on the downstream primer element of lsy-6 miRNA to open up the chromatin. (2) The priming event allows boosting of lsy-6 miRNA levels by the CHE-1 transcription factor in the ASE mother cell, leading to high levels of the miRNA in the postmitotic ASE neuron. (3) The lsy-6 miRNA directly inhibits the ASE promoting transcription factor COG-1, and allows the ASE promoting transcription factor DIE-1 to be expressed. (4) DIE-1 then activates ASE effector genes and represses the ASE cell fate.

ASER: (1) Notch is expressed in the ASER precursor and inhibits the lsy-6 priming event. Therefore, the lsy-6 miRNA expression level cannot be boosted as the chromatin remains compact. (2) Consequently, in the postmitotic ASER neuron, the COG-1 transcription factor is expressed, which activates another miRNA, mir-273. (3) mir-273 directly inhibits the ASE promoting transcription factor DIE-1, and therefore promotes the ASER fate.

Figure 2: Model of mir-71 function in AWC asymmetry. In the AWC OFF cell, high calcium level activates the calcium-regulated UNC-43/TIR-1/MAPK cascade, leading to srsx-3 expression. In the AWC ON cell, nsy-4 and nsy-5 promote the stability of mature mir-71 through an unknown mechanism. The mature mir-71 miRNA targets the 3'UTR of tir-1 mRNA for degradation, leading to the inhibition of the calcium signaling pathway and the subsequent induction of str-2 expression. Orange color represents factors that promote AWC ON. Blue color represents factors that promote AWC OFF. Gray/white color represents inactive or less active factors. Dotted line represents factors being downregulated. PD, pentanedione; BU, butanone.

Mentioned biomedical entities:

Genes: ASE, ASER, CHE-1, COG-1, DIE-1, LIN-14, LMO4, Myt1, Notch, TRIM32, dicer, die-1, let-7, lin-4, lsy-6, miR-9a, mir-273, nsy-4, srsx-3, str-2, xrn-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell, neuron, progenitor cell

Organisms: human, humans, mice, zebrafish

Tissues: N/A

Pathways: N/A

Query Result 5/11

A Genome-Wide RNAi Screen for Factors Involved in Neuronal Specification in *Caenorhabditis elegans*, bScreen for Neuronal Fate Mutants

Poole, Richard J.; Bashllari, Enkelejda; Cochella, Luisa; Flowers, Eileen B.; Hobert, Oliver; Mango, Susan E.

PLoS Genetics

Year: 2011 PMID: 21698137 Citations: 33 Score: 19.57 DOI: 10.1371/journal.pgen.1002109

Keywords Hits: lsy-6: 35, che-1: 12

Keywords Frequency (TF): lsy-6: 0.003; che-1: 0.001

Abstract:

One of the central goals of developmental neurobiology is to describe and understand the multi-tiered molecular events that control the progression of a fertilized egg to a terminally differentiated neuron. In the nematode *Caenorhabditis elegans*, the progression from egg to terminally differentiated neuron has been visually traced by lineage analysis. For example, the two gustatory neurons ASEL and ASER, a bilaterally symmetric neuron pair that is functionally lateralized, are generated from a fertilized egg through an invariant sequence of 11 cellular cleavages that occur stereotypically along specific cleavage planes. Molecular events that occur along this developmental pathway are only superficially understood. We take here an unbiased, genome-wide approach to identify genes that may act at any stage to ensure the correct differentiation of ASEL. Screening a genome-wide RNAi library that knocks-down 18,179 genes (94% of the genome), we identified 245 genes that affect the development of the ASEL neuron, such that the neuron is either not generated, its fate is converted to that of another cell, or cells from other lineage branches now adopt ASEL fate. We analyze in detail two factors that we identify from this screen: (1) the proneural gene *hlh-14*, which we find to be bilaterally expressed in the ASEL/R lineages despite their asymmetric lineage origins and which we find is required to generate neurons from several lineage branches including the ASE neurons, and (2) the COMPASS histone methyltransferase complex, which we find to be a critical embryonic inducer of ASEL/R asymmetry, acting upstream of the previously identified miRNA *lsy-6*. Our study represents the first comprehensive, genome-wide analysis of a single neuronal cell fate decision. The results of this analysis provide a starting point for future studies that will eventually lead to a more complete understanding of how individual neuronal cell types are generated from a single-cell embryo.

Summary:

Introduction: RNAi technology in *C.elegans* has opened an alternative path to classic forward analysis to study the genetic bases of various processes. However, this approach has not been undertaken in *C.elegans* to comprehensively study neuronal development and fate determination. We are using the ASE gustatory neuron class, composed of two bilaterally symmetric neurons, ASEL and ASER, as a paradigm to understand how neuronal fate is specified. Each neuron is produced by two distinct descendants of the embryonic blastomere AB called ABa (generates ASEL) and ABp (generates ASER). There is a progressive restriction in the types of cells produced by the AB blastomere. The great-granddaughter of AB, ABalp, still produces ASEL.

Results: RNAi is known to work less efficiently in the nervous system than in other tissues. Indeed, we find that dsRNA directed against *gfp* barely reduces *gfp* expression levels in transgenic animals that drive *agfp* reporter in the ASEL neuron. Several genetic backgrounds have been described in which animals are more sensitive to RNAi sensitized. We find that in all of them ASE-specific *gfp* population is now significantly reduced, with *nre-1*, *lin-15*, *bander-1*; *li-15* mutants allowing more effective *gfp* knockdown than *rrf-3*. 10.1371/journal.pgen.1002109.g002 Figure 2 Testing hypersensitive strains and validating the screen with visible phenotypes.

Discussion: We describe the first comprehensive, genome-wide RNAi screen that monitors a single neuron-type cell fate decision. The screen has been remarkably successful in retrieving genes known to be involved in controlling ASE neuron specification. Six genes previously known to be involved in ASE Neuron Specification have been found in an unbiased manner (*che-1*, *die-1*, *lsy-2*, *lim-6*, *fozi-1* and only two have not (*cog-1*, *nhr-67*: these genes also display no laterality defects with sequence-verified RNAi clones). The RNAi screen also revealed a number of genes that we would have predicted to result in ASA lineage defects, including genes involved in the segregation of early patterning cues (e.g. *parandmex*-genes) and involved in binary fate decision system (*lit-1* gene).

Figures:

Figure 1: The embryonic origins and specification of the ASE neurons.

Figure 2: Testing hypersensitive strains and validating the screen with visible phenotypes.

Figure 3: Overview of the screening strategy.

Figure 4: Examples of phenotypic classes uncovered by the screen.

Figure 5: The achaete-scute homolog *hlh-14* is required to specify neurons in the ASE lineage.

Figure 6: Asymmetric expression and function of *hlh-14* in the C-lineage.

Figure 7: The evolutionary conserved methyltransferase COMPASS complex is required for ASE asymmetric cell fate specification.

Figure 8: The cloning and expression of *Isy-15/rbbp-5(ot86)* .

Mentioned biomedical entities:

Genes: ASE, ASER, Brd, Gro, LIN-49, Lon, MYST, NBP, Notch, P0, P3, PHD, RbBP5, Unc, WD40, WDR5, WDR82, ash-2, cog-1, die-1, dpy-7, foz-1, histone, lim-6, lin-32, Isy-6, mex-5, nhr-67, pie-1, ref-1, sys-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell, ectoderm, muscle cell, neuron

Organisms: C. elegans, C.elegans, human

Tissues: blastomere, cholinergic neuron, cuticle, egg, neuron, tissue, ventral

Pathways: N/A

Query Result 6/11**Evolution of neuronal anatomy and circuitry in two highly divergent nematode species**

Hong, Ray L; Riebesell, Metta; Bumbarger, Daniel J; Cook, Steven J; Carstensen, Heather R; Sarpolaki, Tahmineh; Cochella, Luisa; Castrejon, Jessica; Moreno, Eduardo; Sieriebriennikov, Bogdan; Hobert, Oliver; Sommer, Ralf J; Sengupta, Piali; Marder, Eve; Sengupta, Piali

eLife

Year: 2019 PMID: 31526477 Citations: 35 Score: 15.88 DOI: 10.7554/eLife.47155.053

Keywords Hits: Isy-6: 4, che-1: 33

Keywords Frequency (TF): Isy-6: 0.000; che-1: 0.002

Abstract:

The nematodes *C. elegans* and *P. pacificus* populate diverse habitats and display distinct patterns of behavior. To understand how their nervous systems have diverged, we undertook a detailed examination of the neuroanatomy of the chemosensory system of *P. pacificus*. Using independent features such as cell body position, axon projections and lipophilic dye uptake, we have assigned homologies between the amphid neurons, their first-layer interneurons, and several internal receptor neurons of *P. pacificus* and *C. elegans*. We found that neuronal number and soma position are highly conserved. However, the morphological elaborations of several amphid cilia are different between them, most notably in the absence of winged cilia morphology in *P. pacificus*. We established a synaptic wiring diagram of amphid sensory neurons and amphid interneurons in *P. pacificus* and found striking patterns of conservation and divergence in connectivity relative to *C. elegans*, but very little changes in relative neighborhood of neuronal processes. These findings demonstrate the existence of several constraints in patterning the nervous system and suggest that major substrates for evolutionary novelty lie in the alterations of dendritic structures and synaptic connectivity.

Summary:

Introduction: Comparative studies on nervous system anatomy have a long tradition of offering fundamental insights into the evolution of nervous systems and, consequently, the evolution of behavior. Traditionally, such comparative studies have relied on relatively coarse anatomical and morphometric comparisons. The relative simplicity of nematode nervous systems facilitates the determination and subsequent comparison of neuroanatomical features of distinct nematode species, thereby enabling an understanding of how members of the same phylum, sharing a common body plan, can engage in very distinct behaviors.

Results: The amphid sensory circuitry of *P. pacificus* is characterized by a combination of 3D reconstructions from TEM sections of two young adult hermaphrodites, as well as live dye uptake and transgene reporter analysis. The amphid neuronal network is similar to that of *C. elegans*. The cell bodies of all amphid neurons and the amphid sheath cell (AMsh) are located in the lateral ganglion posterior to the nerve ring.

Discussion: In the absence of morphological characteristics such as the winged endings of AWx neurons in *C. elegans* we proposed neuronal homologies between *P. pacificus* and *C. elegans* amphid neurons based on their relative cell body positions, axon projections, the manner of dendrite entry into the amphid sheath cell, the number of cilia in channel, dye filling properties, and the connections to first layer interneurons.

Figures:

Figure 1: EM of the amphid sensillum and two other sensory neurons of *P. pacificus* hermaphrodite adults.

Figure 2: URX is ciliated.

Figure 3: URY, URX, URA and BAG endings.

Figure 4: A comparison of select non-amphid ciliated neurons in three nematode species.

Figure 5: Video of URX 3D reconstruction.

Figure 6: Video of URY 3D reconstruction.

Figure 7: Video of URA 3D reconstruction.

Figure 8: Position of amphid cilia in AMso and AMsh.

Figure 9: Posterior-anterior path of amphid dendrite entry into the left sheath glia of specimen 107.

Figure 10: Posterior-anterior path of amphid dendrite entry into the right sheath glia of specimen 107.

Figure 11: Overview of the amphid sensilla.

Figure 12: 3D rendering of AM1(ASH) neurons.

Figure 13: 3D rendering of AM2(ADL) neurons.

Figure 14: 3D rendering of AM3(AWA) neurons.

Figure 15: 3D rendering of AM4(ASK) neurons.

Figure 16: 3D rendering of AM5(ASE) neurons.

Figure 17: 3D rendering of AM6(ASG) neurons.

Figure 18: 3D rendering of AM7(AWC) neurons.

Figure 19: 3D rendering of AM8(ASJ) neurons.

Figure 20: 3D rendering of AM9(ADF) neurons.

Figure 21: 3D rendering of AM10(ASI) neurons.

Figure 22: 3D rendering of AM11(AWB) neurons.

Figure 23: 3D rendering of AM12(AFD) neurons.

Figure 24: 3D rendering of AMU1(AUA) neurons.

Figure 25: 3D rendering of Amphid sheath (AMsh).

Figure 26: 3D rendering of Amphid socket (AMso).

Figure 27: Dye filling and reporter gene expression in *P.pacificus* amphid neurons.

Figure 28: Receptor-type guanylyl cyclases in *P.pacificus* (red font) and *Caenorhabditis* species (black font).

Figure 29: Abbreviated phylogeny of ASE taste receptor-type guanylyl cyclases in *P. pacificus* (red font) and *Caenorhabditis* species (black font).

Figure 30: Ppa-odr-7p::Cel-oig-8 mis-expression.

Figure 31: Orthology assignments for che-1 , odr-7 and odr-3.

Figure 32: Single molecule FISH of Ppa-che-1.

Figure 33: Z-stack images of Dil stained J3 larva (ventral view).

Figure 34: Z-stack images of Ppa-che-1p::rfp stained with DiO.

Figure 35: Z-stack images of Ppa-odr-7p::rfp .

Figure 36: Comparison of the reconstructed cilia of *P. pacificus* and *C. elegans* amphid neurons.

Figure 37: Genetic marker for the amphid glia.

Figure 38: 3D rendering of Ppa-daf-6::rfp J4 with confocal microscopy.

Figure 39: Comparison of cuticular sensilla and internal receptors.

Figure 40: The first layer amphid interneurons.

Figure 41: Comparisons of synaptic connectivity.

Mentioned biomedical entities:

Genes: ADF, ADL, AM11, AM5, AM6, ASE, ASER, ASK, B, BAG, CHE-1, DAF-6, FLP, FP, GA, PHB, PPA01143, Ppa-nhr-66, Ppa-rpl-23, URB, VG, gcy, gcy-6, lsy-6, odr-7, str-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell, epidermal cell, neuron, neurotransmitter, secretory cell, sheath cell

Organisms: *A. complexus*, *Acrobeles complexus*, *C. briggsae*, *C. elegans*, *Drosophila melanogaster*, *E. coli*, *H. contortus*, *Haemonchus*, *Haemonchus contortus*, *Homo sapiens*, *Oesophagostomum dentatum*, *Onchocerca volvulus*, *Rhabditidae*

Tissues: cuticle, dorsal, neuron, tubular, ventral

Pathways: N/A

Query Result 7/11

Single-cell transcriptional analysis of taste sensory neuron pair in *Caenorhabditis elegans*

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Nucleic Acids Research

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Abstract:

The nervous system is composed of a wide variety of neurons. A description of the transcriptional profiles of each neuron would yield enormous information about the molecular mechanisms that define morphological or functional characteristics. Here we show that RNA isolation from single neurons is feasible by using an optimized mRNA tagging method. This method extracts transcripts in the target cells by co-immunoprecipitation of the complexes of RNA and epitope-tagged poly(A) binding protein expressed specifically in the cells. With this method and genome-wide microarray, we compared the transcriptional profiles of two functionally different neurons in the main *C. elegans* gustatory neuron class ASE. Eight of the 13 known subtype-specific genes were successfully detected. Additionally, we identified nine novel genes including a receptor guanylyl cyclase, secreted proteins, a TRPC channel and uncharacterized genes conserved among nematodes, suggesting the two neurons are substantially different than previously thought. The expression of these novel genes was controlled by the previously known regulatory network for subtype differentiation. We also describe unique motif organization within individual gene groups classified by the expression patterns in ASE. Our study paves the way to the complete catalog of the expression profiles of individual *C. elegans* neurons.

Summary:

INTRODUCTION: Each neuron in a mature nervous system has various physiological, biochemical and morphological properties, which are defined by the specific set of expressed genes in the neurons. Therefore, a description of the transcriptional profiles of each neuron would provide a basis for understanding the molecular mechanisms that generate neuronal diversity. With its relatively simple nervous system and fully-sequenced genome, the model nematode *Caenorhabditis elegans* offers a unique opportunity to attempt this challenging research objective).

RESULTS: Optimizing the mRNA tagging/microarray profiling method for single cells. To enrich for transcripts from the single mature taste neurons, we generated two *C. elegans* lines stably expressing FLAG PABP in ASE or ASER from chromosomally integrated transgenes. FLAGPABP was expressed by the promoter of *gcy-7* or *gcy-5*, which drives expression exclusively in ASER or ASE, respectively. Single cell expression of FLAG APBP was confirmed by immunohistochemistry with the anti-FLAG antibody. The efficiency of enrichment was assessed by quantifying the relative amount of the transcripts in the extracted RNA samples by real-time quantitative RT-PCR.

DISCUSSION: This study shows that transcript extraction from single cells is feasible by a modified mRNA tagging method. This technique can be used in applications from single cell to large tissues, so long as an appropriate promoter is available. The major advantages of this method are its independence of any regulatory factors and applicability to every developmental stage, while limitation to generality of this methodology would be the requirement of appropriate cell-specific promoters. Since the transcriptional profiling of a small number of cells in *C. elegans* was so far limited to embryonic cells, our study opened the door to the impartial expression profiling of single cells that differentiate or are even generated post-embryonically.

Figures:

Figure 1: FLAG::PABP is expressed in single neurons of the transgenic animals. Expression of FLAG::PABP in the single cells of transgenic worms carrying ASE_{LP}::flag::pabp (A) and ASER_P::flag::pabp (B) are visualized by immunohistochemistry with anti-FLAG antibodies. Arrowheads point to the immunofluorescence signals. Top panels are immunofluorescence images, and bottom panels are differential interference contrast (DIC) images. Scale bar = 25 μ m.

Figure 2: Representative fluorescence images for novel ASER-biased genes. (A) Schematic depiction of the locations of ASE neurons. Circles indicate the ASE and ASER neurons. (B-H) Transgenic animals carrying promoter-fused Venus reporter genes. Animals at the L1 stage (B-F) and the adult stage (G, H) are shown. Scale bar = 10 μ m.

Figure 3: Quantification of the expression bias of the newly identified genes in wild-type and mutant backgrounds. (A) Lateralized gene expression in ASE is controlled by the upstream double-negative feedback loop and the downstream transcription factors (6). ASER-biased *gcy* genes are regulated by *lim-6*, whereas *hen-1* is independent of *lim-6*. Genes in active or inactive state are shown in black or grey, respectively. (B) Quantification of the expression laterality in ASE neurons of the newly identified genes in wild-type and the mutant backgrounds. The expression bias is indicated as follows except for *trp-2*: L 0 and 0 R refer to the restricted expression to ASE or ASER, respectively; L R and L R refer to different levels of expression in ASE versus ASER; L = R refers to equal expression in ASE and ASER; 0 = 0 refers to no expression in ASE nor in ASER. For *trp-2*, since strong Venus expression in *trp-2p::Venus* worms were seen in neurons just medial to ASE, the expression was observed on one side

for each animal. Strong, Weak and None refer to the expression levels at each neuron. The number of examined animals is shown in each graph.

Figure 4: The newly identified genes with ASER-biased expression have ASE motifs. All of the ASER-biased genes had ASE motifs in the promoter region, whereas ASEL-biased ins-32 did not. The location of ASE motifs was shown by arrow heads. Notably, six of the eight ASER-biased genes harbored multiple motifs within the promoter region.

Figure 5: Similar ASE motif organization was observed within each gene set classified by the expression laterality. (A) The occurrence frequency of ASE motifs was high in the genes expressed in ASE neurons. ASEL, Bilateral, ASER and Neither refer to the gene set with corresponding expression in ASE neurons. Genes in the Random set were selected randomly from gene spots in the microarray. (B) Most of the ASEL-biased genes had single ASE motifs, whereas many of the other two gene sets had multiple motifs ($P = 0.05$; randomization test). Bars refer to the average number of motifs. Genes with no ASE motif were discarded, since they were assumed to be indirectly regulated by CHE-1. (C) No strong correlation was observed between the length of a promoter and the number of motifs per promoter. (D) A was most frequently observed at the fourth nucleotide within the ASE motifs of the promoters with ASEL-biased expression, whereas G was the majority in genes expressed in ASER. (E) T was most frequently observed at the seventh nucleotide in ASEL-biased promoters, whereas no strong preference in nucleotide usage was seen in the promoters with equivalent or ASER-biased expression. (F) Expression laterality in ASE neurons of wild-type nlp-5 promoter and mutant promoter that has ASEL-type mutations in ASE motif. Laterality is presented in the same manner as Figure 3 .

Mentioned biomedical entities:

Genes: 27, ASE, ASER, Bad, C, CHE-1, PABP, Rex, T, TRPC, cog-1, die-1, dpy-20, flp, foz-1, gcy-22, gcy-5, gcy-6, lim-6, lsy-6, nlp-5, nlp-7

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *C. elegans*

Tissues: body wall muscle, chemoreceptor, neuron

Pathways: N/A

Query Result 8/11**MicroRNAs: Not Fine-Tuners but Key Regulators of Neuronal Development and Function**

Davis, Gregory M.; Haas, Matilda A.; Pocock, Roger

Frontiers in Neurology

Year: 2015 PMID: 26635721 Citations: 42 Score: 8.26 DOI: 10.3389/fneur.2015.00245

Keywords Hits: lsy-6: 7, che-1: 1

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.000

Abstract:

MicroRNAs (miRNAs) are a class of short non-coding RNAs that operate as prominent post-transcriptional regulators of eukaryotic gene expression. miRNAs are abundantly expressed in the brain of most animals and exert diverse roles. The anatomical and functional complexity of the brain requires the precise coordination of multilayered gene regulatory networks. The flexibility, speed, and reversibility of miRNA function provide precise temporal and spatial gene regulatory capabilities that are crucial for the correct functioning of the brain. Studies have shown that the underlying molecular mechanisms controlled by miRNAs in the nervous systems of invertebrate and vertebrate models are remarkably conserved in humans. We endeavor to provide insight into the roles of miRNAs in the nervous systems of these model organisms and discuss how such information may be used to inform regarding diseases of the human brain.

Summary:

Introduction: MicroRNAs (miRNAs) are non-coding RNA molecules with a length of approximately 22 nucleotides, which act as post-transcriptional regulators of gene expression. Discovered just over two decades ago, miRNAs have been found to be abundant in most organisms and critically important for post-transcriptional control of mRNAs by regulating a predicted 60 of protein-coding genes. Prior to the discovery of the miRNA pathway, the *lin-14* gene in *Caenorhabditis elegans* was shown to be regulated by a 22-nucleotide partially complementary strand of RNA called *lin-4*. However, a mechanistic understanding of this process remained unclear until the *let-7* gene was shown to encode a complementary sequence of *lin-4* to regulate developmental timing. This led to a paradigm shift in how mRNA regulation was viewed, and further investigation demonstrated that the miRNA pathway was evolutionarily conserved in most eukaryotes. Since then miRNAs have been shown to have been required for key biological processes, such as cell fate, differentiation, apoptosis, and tumor suppression ().

Conclusion: *lsey-6* miRNA is a small non-coding RNA that is involved in the nervous system. It is regulated by the *lin-4* and *let-7* miRNAs. These miRNAs are involved in a number of biological processes, including neuronal development, neuron differentiation, and neuron function. These non-coded RNAs are regulated in the brain by the *miR-44* and *miR-4* miRNA. These RNAs are a part of the mRNA family.

Figures:

Figure 1: The miRNA pathway. Primary miRNAs (pri-miRNAs) are transcribed from the genome and form hairpin structures. Nuclear-localized Drosha endonuclease cleaves pri-miRNAs into approximately 70nt precursor miRNAs (pre-miRNAs) which are then transported from the nucleus to the cytoplasm by Exportin-5 (XPO5) via the nuclear pore complex, where they are further cleaved by Dicer into mature 21-23nt miRNA fragments. Once the strands separate, the guide strand is loaded into the RISC complex (AGO and different cofactors) to scan the transcriptome for partial complementary target transcripts. These sequences are either repressed by the RISC complex or degraded in P-bodies.

Figure 2: Roles of miRNAs in different stages of neuronal development. miRNAs are involved in the multiple stages of neuronal development in invertebrates and vertebrates. Listed here are the miRNAs we cover in this review that regulate single or multiple stages of neuronal development. *cel*, *Caenorhabditis elegans*; *dme*, *Drosophila melanogaster*; *dre*, *Danio rerio*; *hsa*, *Homo sapiens*; *mmu*, *Mus musculus*; *xtr*, *Xenopus tropicalis*.

Mentioned biomedical entities:

Genes: AGO, ASCL1, Akt, Argonaute, CHE-1, COG-1, CamkII, Cxcr4b, DAF-16, Dgcr8, Dicer, EGFR, Emx1, ErbB2, FMR1, FOXO, Fgf, Fgf8, Fmr1, Foxg1, Foxp2, Gsh2, Hes1, LIN-14, LON-2, Let-7, Lmx1b, MAP2, MECP2, MEF-2, MLN51, MYT1L, MeCP2, NEUROD2, NRXN1, Notch, Nrg, Phf6, REST, RhoG, SHANK3, SOX-2, SQV-7, Sox9, Syt1, TLX, TRIM, TRIM32, UNC-40, Wnt, XPO5, *ana*, *chinmo*, *cnot8*, *dicer*, *dme*, *fgfr1*, *her5*, *jigr1*, *let-7*, *lin-14*, *lin-28*, *lin-4*, *lsey-6*, *miR-124*, *miR-125*, *miR-125b*, *miR-137*, *miR-228*, *miR-289*, *miR-8*, *miR-9*, *miR-92*, *miR-92b*, *miR-9a*, *mir-124*, *mir-71*, *pitx3*, *zic5*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, Neuron, cell, glial cell, neural progenitor cell, neural stem cell, neuron, neurotransmitter, stem cell

Organisms: *Danio rerio*, *Drosophila melanogaster*, *Homo sapiens*, *Mus musculus*, human, humans, mice, mouse, zebrafish

Tissues: epidermis, neuron, synapse, tissue

Pathways: N/A

Query Result 9/11**Cell-type specific sequencing of microRNAs from complex animal tissues**

Alberti, Chiara; Manzenreither, Raphael A.; Sowemimo, Ivica; Burkard, Thomas R.; Wang, Jingkui; Mahofsky, Katharina; Ameres, Stefan L.; Cochella, Luisa

Nature methods

Year: 2018 PMID: 29481550 Citations: 51 Score: 7.60 DOI: 10.1038/nmeth.4610

Keywords Hits: lsy-6: 5, che-1: 2

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.000

Abstract:

MicroRNAs (miRNAs) play an essential role in the post-transcriptional regulation of animal development and physiology. However, in vivo studies aimed at linking miRNA-function to the biology of distinct cell types within complex tissues remain challenging, partly because in vivo miRNA-profiling methods lack cellular resolution. We report microRNome by methylation-dependent sequencing (mime-seq), an in vivo enzymatic small RNA-tagging approach that enables high-throughput sequencing of tissue- and cell-type-specific miRNAs in animals. The method combines cell-type-specific 3-terminal 2-O-methylation of animal miRNAs by a genetically encoded, plant-specific methyltransferase (HEN1), with chemoselective small RNA cloning and high-throughput sequencing. We show that mime-seq uncovers the miRNomes of specific cells within *C. elegans* and *Drosophila* at unprecedented specificity and sensitivity, enabling miRNA profiling with single-cell resolution in whole animals. Mime-seq overcomes current challenges in cell-type-specific small RNA profiling and provides novel entry points for understanding the function of miRNAs in spatially restricted physiological settings.

Summary:

Introduction: Post-transcriptional silencing by small regulatory RNAs such as miRNAs play essential roles in shaping gene expression during these processes. These 21-24-nt long RNA act in an RNA-induced silencing complex (RISC) with an Argonaute-family protein, directing repression of target mRNAs through base-pairing interactions typically between the mRNA 3UTR and the miRNA 5' end or seed region. The functional contribution of miRNA, or other gene expression regulators, to the biology of an organism depends on the cells in which they act. Thus, miRNA profiling methods are powerful entry points to dissect the roles of members of this repressor class.

Results: An Arabidopsis RNA methyltransferase HEN1 methylates animal miRNAs. RNA-methylation is a common modification of small RNAs in plants and animals. This modification is absent from the bulk of animal miRNAs. We show that methylation of RNA is regulated by the RNA polymerase Ath-HEN1 in *Drosophila* S2 cells. This methylated RNA preserves an adapter-ligatable 3'OH for cDNA library preparation.

Discussion: Mime-seq overcomes current challenges for in vivo miRNA profiling from specific cell types within complex tissues or whole organisms. It circumvents excessive manipulations such as cell sorting or immunoprecipitation, which reduce yields and increase noise, achieving greater sensitivity. In direct comparison with neuronal and intestinal Argonaute immunoprecipitations in *C. elegans*, mime-seq identified 3- and 2.6-fold more miRNAs respectively, many of which we validated, including a miRNA expressed in a single neuron out of the whole animal.

Figures:

Figure 1: The methyltransferase HEN1 from Arabidopsis thaliana methylates animal miRNAs in vitro and in vivo .

Figure 2: Periodate-mediated oxidation enables selective cloning of animal miRNAs upon restrictive Ath-HEN1-directed methylation.

Figure 3: Mime-seq reproducibly recovers neuronal miRNAs in *C. elegans* .

Figure 4: Mime-seq unveils the miRNomes of diverse tissues in *C. elegans*.

Figure 5: Mime-seq has the sensitivity to retrieve miRNAs from 1/300 cells.

Mentioned biomedical entities:

Genes: ASE, Actin, Argonaute, Cas9, Gal4, HEN1, HENN-1, Hen1, RDE-1, Tubulin, Unc, attB, dme, elt-2, hen1, lsy-6, miR-252, miR-277, myo-2, rab-3, rgef-1, tRNAGly

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, S2 cell, cell, neuron

Organisms: Mus musculus, flies, mouse

Tissues: body wall muscle, cuticle, seed, tissue, tube

Pathways: N/A

Query Result 10/11**Neurogenesis in *Caenorhabditis elegans***

Poole, Richard J; Flames, Nuria; Cochella, Luisa; Sundaram, M

Genetics

Year: 2024

PMID: 39167071

Citations: 11

Score: 6.30

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Keywords Hits: lsy-6: 17, che-1: 34

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.001

Abstract:

Animals rely on their nervous systems to process sensory inputs, integrate these with internal signals, and produce behavioral outputs. This is enabled by the highly specialized morphologies and functions of neurons. Neuronal cells share multiple structural and physiological features, but they also come in a large diversity of types or classes that give the nervous system its broad range of functions and plasticity. This diversity, first recognized over a century ago, spurred classification efforts based on morphology, function, and molecular criteria. *Caenorhabditis elegans*, with its precisely mapped nervous system at the anatomical level, an extensive molecular description of most of its neurons, and its genetic amenability, has been a prime model for understanding how neurons develop and diversify at a mechanistic level. Here, we review the gene regulatory mechanisms driving neurogenesis and the diversification of neuron classes and subclasses in *C. elegans*. We discuss our current understanding of the specification of neuronal progenitors and their differentiation in terms of the transcription factors involved and ensuing changes in gene expression and chromatin landscape. The central theme that has emerged is that the identity of a neuron is defined by modules of gene batteries that are under control of parallel yet interconnected regulatory mechanisms. We focus on how, to achieve these terminal identities, cells integrate information along their developmental lineages. Moreover, we discuss how neurons are diversified postembryonically in a time-, genetic sex-, and activity-dependent manner. Finally, we discuss how the understanding of neuronal development can provide insights into the evolution of neuronal diversity.

Summary:

Introduction: The complex functionalities of animal nervous systems rely on a vast diversity of neurons, which receive sensory inputs, integrate different internal and external signals, and produce cognitive and behavioral outputs. Neurons are highly specialized cells, with unique morphological and physiological properties and an outstanding capacity for intercellular communication. While they share several common functional and molecular properties, they present in a multitude of different types or classes. Cellular diversity within nervous systems was recognized more than 100 years ago when the remarkable range of differences in neuronal morphologies was revealed and prompted multiple efforts toward neuronal classification, initially based on morphology but later including functional and m

Concluding remarks: The last 2 decades of studies on neuron specification and differentiation of the complete nervous system of *C. elegans* have allowed us to start building a holistic model of the regulatory networks involved in neurogenesis, with a degree of molecular and cellular resolution that is difficult to attain in other animal models. We have distilled the vast genetic and molecule analyses generated by the *C. elegans* oncology, described here, into the regulatory framework summarized in. This view also highlights open questions.

Figures:

Figure 1: Regulatory framework for *C. elegans* neurogenesis. Summary of the main regulatory interactions controlling neurogenesis and neuron diversification that should serve as a roadmap for this chapter. Development of each neuron class requires integration of regulatory information from different developmental timepoints into parallel, yet interconnected regulatory modules that coexist in the cell. (1) Lineage history determines the set of TFs expressed in the neuron and its progenitors, the signalling events it receives over time, and the particular chromatin landscape of the postmitotic neuron (see Lineage-based mechanisms of neuronal diversification). (2) bHLH proneural factors have evolutionary conserved roles in the specification of neuronal progenitors (see Diverse functions of bHLH proneural TFs). (3) Terminal selector TFs act in the postmitotic neuron to directly activate broad neuron class effector gene batteries (see Rich sets of effector genes distinguish different neuron classes and Principles of neuron class specification by terminal selectors). (4) Terminal selectors act together with HOX TFs to diversify some neuron subclasses along the AP axis (see Neuronal diversification across body axes). (5) Neuron class transcriptomes are also shaped by postembryonic time or genetic sex (see Neuronal diversification over developmental time and across sexes). (6) Mature neurons can modify their transcriptome in response to environmental stimuli (see Environmental effects on neuron gene expression). (7 and 8) At least 2 regulatory modules run in parallel to those specifying neuron class properties: panneuronal effector genes, those shared by all neuron classes are under direct control of CUT HD TFs (7) (see Panneuronal features define neurons as a tissue type) while different TFs activate the ciliome components expressed by all sensory ciliated neurons (8) (see Rich sets of effector genes distinguish different neuron classes). t.g., neuron type effector gene; p.g., panneuronal effector gene; c.g., ciliome effector gene.

Figure 2: Developmental origin of neurons and neuronal diversity. a) Matrix of the 118 morphologically different neuron classes in hermaphrodite *C. elegans* classified according to different criteria including embryonic/postembryonic lineage, function, and neurotransmitter identity. For simplicity, large effector gene categories, such as neuropeptides or innexins, have not been included but references to their expression patterns are provided in

the text. b) Lineage origins of all 302 embryonically and postembryonically generated hermaphrodite neurons. c) A cartoon illustration of a generic neuron depicting different sets of proteins required for specific neuronal functions, which include panneuronal factors such as synaptic components, cilium proteins (ciliome) present in ciliated sensory neurons, and different sets of neuron class-specific effector proteins such as adhesion molecules, enzymes, ion channels, receptors, or neuropeptides.

Figure 3: Progressive determination model of neuronal development. A cartoon illustration of the progressive determination model of neural development. Key steps relevant to *C. elegans* include the specification of neural progenitors and the differentiation of postmitotic cells into neurons.

Figure 4: Expression of bHLH TFs in neuronal lineages. a) Summary of proneural gene expression at any point in the lineage of the indicated embryonic neurons (sources: WormAtlas, Sulston and Horvitz 1977 ; Sulston et al. 1983 ; Reilly et al. 2022 ; Masoudi et al. 2023). b) An illustrative example of sequential bHLH expression in the ABalpp/ABpraa lineage. *hlh-3* is expressed both early and broadly and has no known role in this lineage. *hlh-14* is expressed in a more restricted manner and is required for neural precursor specification. *hlh-4* is only expressed in the ADL neurons where it acts as a terminal selector and not as proneural gene. See the main text for more details.

Figure 5: Terminal selector functions in neuronal diversification. a) Example of the accumulation of regulatory factors along a lineage to result in specific (and robust) patterns of gene expression (from Walton et al. 2015). b) Terminal selector TFs act combinatorially on modular cis- regulatory sequences, enabling cell-specific control of various terminal effector genes. The modular architecture of cis- regulatory elements enables integration of distinct combinations of terminal selectors, as well as integration with other transcriptional inputs as discussed in the rest of the chapter. The 3 illustrated examples correspond to those described in the text. c) The output of a terminal selector (e.g. *UNC-86*) can be diversified by integration into gene regulatory networks with other transcriptional regulators. These can be organized in regulatory motifs such as coherent feedforward loops (in which a terminal selector activates an effector gene as well as an additional transcriptional activator, blue arrows) and incoherent feedforward loops (in which a terminal selector activates an effector gene as well as a repressor of the same target gene, red arrows). Such motifs are recurrently used to modify the terminal battery activated by a terminal selector. d) Regulation of the different components of the neuronal transcriptome is achieved thanks to combinatorial TF action on modular cis- regulatory sequences that result in parallel modules that are interconnected. For example, panneuronal gene expression relies on CUT TFs but also receives terminal selector input. Gene modules such as those coding for the structural cilium components are under control of a parallel program driven by *FKH-8* and *DAF-19C*, but other neuron class-specific components of the cilia, just like other neuron-type effector genes, are under terminal selector control.

Figure 6: Generation of terminal selector combinations in space and time. a) Schematic of how the Wnt asymmetry pathway can be integrated with existing TFs to generate a novel regulatory state during every cell division (adapted from Bertrand 2016). The asymmetry in *POP-1* and *SYS-1* results in distinct transcriptional outputs in the anterior vs the posterior daughter cell. Note that this schematic is highly simplified and *POP-1* in the anterior cell can also activate or repress targets (Murgan and Bertrand 2015). b) Specification of the AIY neurons illustrates a number of points described in the text. The vertical axis denotes time. Transient expression of a trigger, *REF-2*, is integrated with the Wnt asymmetry pathway and autoactivation of the terminal selectors *TTX-3* and *CEH-10* to result in specific and robust activation in the posterior cell. This distinguishes the SMDD (anterior) from the AIY (posterior) neurons. c) The autoactivation mechanism of *che-1* has been proposed to be sensitive to fluctuations in the *CHE-1* protein by preferential activation of its own locus (Traets et al. 2021). d) The differentiation of the left and right ASE neurons also illustrates a number of mechanisms described in the text. These 2 neurons derive from lineages that diverge at the 4-cell stage owing to the first Notch induction and represent one of the best described examples of convergence. The point where the lineages converge is marked with a dashed line. The converging branches generate 11 pairs of bilaterally symmetric neurons and 2 pairs of bilaterally symmetric glia. The early lineage asymmetry is integrated with the terminal selector *CHE-1* in the form of differential chromatin states of the *Isy-6* locus. This determines whether *Isy-6* is expressed (ASEL) or not (ASER) and the downstream acquisition of the respective terminal gene batteries (Hobert 2014) (Fig. 7a).

Figure 7: Repressor-based mechanisms for neuronal diversification. a) Schematic of the double-repressor interactions that result in ASE asymmetry. The miRNA *Isy-6* in ASEL represses the production of the TF *COG-1*. *cog-1* is transcribed in both ASEs but only produces *COG-1* protein in ASER where *Isy-6* is absent. *COG-1* acts with the corepressor *Groucho/UNC-37* to repress transcription of *die-1*, which encodes another TF. The mutually exclusive expression of *COG-1* and *DIE-1* define downstream molecular asymmetries that determine the asymmetric sensory function of the 2 ASEs. b) Scheme of the repressor-based diversification of distinct motor neuron classes. All classes rely on the terminal selector *UNC-3* for their differentiation, but each class expresses a distinct combination of repressors that counteract the action of *UNC-3* on specific gene batteries (Kerk et al. 2017). c) The action of the terminal selector *UNC-86* in the ALM and BDU sister neurons is diversified by the asymmetric expression of *MEC-3* in ALM. In the absence of *MEC-3*, *UNC-86* acts cooperatively with *PAG-3* to activate BDU-specific gene expression. In ALM, *MEC-3* binds *UNC-86*, which now activates the ALM-specific battery and is titrated away from *PAG-3*-dependent genes (Gordon and Hobert 2015). d) The miRNA *miR-791* is specifically expressed in the BAG, AFD, and ASE neurons, where it posttranscriptionally represses 2 otherwise ubiquitously transcribed genes, *cah-3* and *akap-1*. This repression is necessary in the BAG neurons to elicit the correct avoidance response to high CO₂ levels (Drexel et al. 2016). e) Key components of the heterochronic pathway and their regulatory interactions (Ambros 2000 ; Ketting and Cochella 2021). Proteins are in blue; regulatory RNAs are in red and include the miRNAs *lin-4* and *let-7* and the *lep-5* lncRNA that associates with *Makorin/LEP-2* (Herrera et al. 2016 ;

Kiontke et al. 2019). LIN-14, HBL-1, and LIN-29 are TF outputs of the pathway that are integrated with other TFs (e.g. terminal selectors) on the cis-regulatory regions of neuronal genes discussed in the text.

Figure 8: Environmental effects on neuron diversity. a) Different environmental triggers are integrated into gene expression changes through parallel modules acting in the neuron. b) L4 expression data of signal activated TFs and some examples of terminal selectors in sensory neurons. Signal-activated TFs, such as MEF-2, CRH-1, and DAF-16, are broadly expressed in the nervous system; however, they produce neuron class-specific responses acting together with terminal selectors that show a more restricted pattern of expression. Data source CENGEN. c) The 2 illustrated examples correspond to those described in the text. CRMs of plastic effector genes can integrate input from signal-regulated TFs as well as terminal selectors. ADL expression of *srh-234* is directly regulated by the HLH-4 terminal selector but upon starvation is repressed by the MEF-2 TF. *tph-1* expression in the ADF neuron is plastic to several environmental conditions, and pathogen exposure induces *tph-1* expression through the activity of the LAG-1 ADF terminal selector.

Mentioned biomedical entities:

Genes: ADF, ALN, ASE, ASER, AST-1, B, BAG, BNC-1, CEH-10, CEH-13, CEH-14, CEH-20, CEH-34, CEH-36, CHE-1, COG-1, CRH-1, CUT, CtBP, DA, DAF-16, DAF-19, DAs, DB, DDs, DIE-1, DMD-3, DR, Dop, EGL-44, EGL-5, FKH-8, FLP, Glu, Groucho, HES, HLH-2, Hh, I3, LAG-1, LIN-11, LIN-14, LIN-26, LIN-28, LIN-29, LIN-39, LIN-41, LIN-53, MAB-3, MEC-3, MEF-2, MyT1, MyoD, NGF, NHR, Neurogenin, Notch, ONECUT, Olig, P38, PA, PAG-3, PHB, PHP-3, POP-1, REF-1, REF-2, RIM, RIP, SAB, SOX-2, STR-2, SYS-1, Sox, Sox-3, Sox2, SoxB, SoxC, TBX-37, TF, TRA-1, TTX-3, UNC-130, UNC-18, UNC-3, UNC-30, UNC-39, UNC-4, UNC-55, UNC-86, V5, VAB-7, VCs, VD, VNC, Wnt, ZAG-1, *bas-1*, *cah-3*, *ced-3*, *ceh-13*, *ceh-30*, *ceh-36*, *ceh-6*, *ces-1*, *ces-2*, *cog-1*, *die-1*, *egl-44*, *egl-46*, *egl-5*, *ham-1*, *histone*, *hlh-16*, *hlh-17*, *hlh-3*, *hlh-6*, *let-7*, *lin-14*, *lin-26*, *lin-28*, *lin-32*, *lin-39*, *lin-4*, *lsy-6*, *mab-5*, *ngn-1*, *nob-1*, *odr-10*, *php-3*, *sox-2*, *str-2*, *unc-3*, *unc-30*, *unc-39*, *unc-4*, *unc-6*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, GLR cell, cell, glial cell, interneuron, muscle cell, neural progenitor cell, neural stem cell, neuron

Organisms: C. elegans, cnidarians, humans, mice, mouse, zebrafish

Tissues: blastomere, body wall muscle, egg, hypodermis, neural tissue, neuron, synapse, tissue, ventral

Pathways: N/A

Query Result 11/11

Plasticity in gustatory and nociceptive neurons controls decision making in *C. elegans* salt navigation

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Abstract:

A conventional understanding of perception assigns sensory organs the role of capturing the environment. Better sensors result in more accurate encoding of stimuli, allowing for cognitive processing downstream. Here we show that plasticity in sensory neurons mediates a behavioral switch in *C. elegans* between attraction to NaCl in naive animals and avoidance of NaCl in preconditioned animals, called gustatory plasticity. Ca²⁺ imaging in ASE and ASH NaCl sensing neurons reveals multiple cell-autonomous and distributed circuit adaptation mechanisms. A computational model quantitatively accounts for observed behaviors and reveals roles for sensory neurons in the control and modulation of motor behaviors, decision making and navigational strategy. Sensory adaptation dynamically alters the encoding of the environment. Rather than encoding the stimulus directly, therefore, we propose that these *C. elegans* sensors dynamically encode a context-dependent value of the stimulus. Our results demonstrate how adaptive sensory computation can directly control an animals behavioral state.

Summary:

Introduction: Adaptive behavior refers to the ability of animals to change their actions in response to changes in the environment or in their internal state. Here, we study a form of short-term sensory adaptation, called gustatory plasticity. Its defining feature in the nematode *Caenorhabditis elegans* is the dynamic balance of salt (NaCl) attraction and avoidance as a function of experience. Often, the goal of behavior is implicit (e.g., tracking an animal and recording its neural activity may not disclose what form of reward the animal seeks as it acts in its environment). Furthermore, the advantage gained by Adaptive behavior refer to the able of animals a change their behaviors in response t o changes in th e environment or t

Results: The Ca²⁺reporter Yellow Cameleon identifies the range of nave sensory responses to NaCl. These responses are a result of a gradual increase in the fraction of animals that respond to a 3s exposure to different concentrations of NaCr. The Ca²⁺-response index (RI) is a measure of the proportion of animals responding to osmotic stimuli. The response index is derived from the fraction that responded to varying concentrations.

Discussion: In nave *C. elegans*, attractive, ASE mediated, and aversive, ASH mediated salt responses are controlled by clearly delineated subcircuits, resulting in a switch between attraction up to 200mM NaCl and avoidance of higher concentrations (Fig). However, while nave animals are attracted to low salt concentrations, extended NaCr exposure without food leads animals to avoid any NaCc concentration implying an adaptive foraging behavior. Here, we showed that the behavioral switch between NaCla attraction and avoidANCE is mediated by plasticity in sensory neurons, causing altered dynamic ranges in both attractive and nociceptive subcircuit. Based on our experimental and modeling results we propose that the sensory response ofC. eliota to NaCle is regulated at multiple levels.

Reporting summary: The following information is available in the linked to this article: . Further information on research design is available in the linked to this article. . Supplementary information: - . - . - . - . / -- -- : () / - , - , - (-) - / ,

Figures:

Figure 1: Ca²⁺ responses of ASEL, ASER, and ASH neurons to brief NaCl exposure.

Figure 2: Prolonged exposure to NaCl sensitizes ASER.

Figure 3: Pre-exposure to NaCl desensitizes ASEL.

Figure 4: Prolonged exposure to NaCl sensitizes ASH.

Figure 5: ASER sensitization is affected by unc-13 and eat-4 mutations.

Figure 6: ASH sensitization requires ASE, glutamate, neuropeptides, dopamine, and serotonin.

Figure 7: Computational model reproduces sensory responses to NaCl in virtual quadrant assay.

Figure 8: Stochastic recruitment of ASH drives gustatory plasticity in our computational model.

Figure 9: Sensory adaptation controls exploration in a virtual spot assay.

Figure 10: Schematic model of the NaCl navigation circuit.

Mentioned biomedical entities:

Genes: ASE, ASER, C, RI, eat-4, flp-6, gpc-1, lsy-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell, interneuron, neuron

Organisms: N/A

Tissues: capillary, neuron

Pathways: N/A